

PART IB Oscillations, Waves and Optics Notes

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0 Introduction

Oscillations happens at every second. From the start of the universe. They are of great importance, not only because they are useful in engineering applications, but also because they are the fundamental blocks of modern theories of physics. We will see why this is true in the following sections.

The notes are based on the Natural Sciences Part IB Oscillations, Waves and Optics course at the University of Cambridge. Lectures were delivered by Professor Tijmen Euser [1].

I would like to say thank you to my supervisor in this course, Mr Matthew Smith, who was also my Part IA Mathematics B supervisor. His supervision sessions were always full of elegant math, inspiring thoughts, beautiful writings and useful connections. I enjoyed every moment with him.

1 Harmonic Oscillators

Simple harmonic motion is defined by its equation of motion¹,

$$m\ddot{x} + kx = 0, \quad (1)$$

and has solution of form

$$x = A \cos(\omega_0 t + \phi), \quad (2)$$

where $\omega_0 = \sqrt{k/m}$ is the natural frequency of the oscillator.

Assume that we have a damping proportional to the speed of the object², and a driven force, the equation of motion turns into

$$m\ddot{x} + b\dot{x} + kx = F(x). \quad (3)$$

The solution therefore depends on the driving force F . Writing the equation in a more suitable form

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = f(x), \quad (4)$$

¹Or equivalently, by its potential $V \propto x^2$.

²In practical situations, such as a simple pendulum, our assumptions may not work very well. The potential may only be quadratic locally, and the drag force may not be proportional to speed. Therefore we need numerical integration methods to solve the differential equation. These are discussed in the computing questions and projects.

where $\gamma = b/m$ and $f = F/m$. Perhaps defining $\gamma = b/2m$ is a mathematically tidier way, but somehow the lecture decided to ignore that intuition. The quality factor, defined by the phase needed for the amplitude to be reduced by a factor of e , is

$$Q = \frac{\omega_0}{\gamma}. \quad (5)$$

To solve the equation of motion, it is often convenient to write x as the real part of the complex representation $x = \Re(Ae^{i\phi}e^{i\omega t})$. This is because it is easy to differentiate and integrate, plus, easy to deal with phases.

In a phasor diagram, position, velocity and acceleration have 90 degrees between them, as shown in Fig. 1

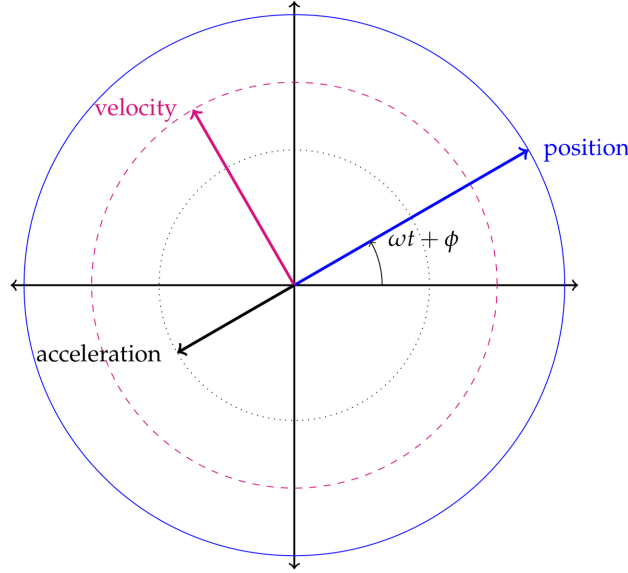


Figure 1: Phasor diagram for position, velocity and acceleration.

With no driving force, the solution to the system depends on the relative value of γ and ω_0 . Note that, γ and ω_0 can be complex mathematically, we're assuming real values only because they are physical quantities. Suppose $x \propto e^{\lambda t}$, λ can be solved as

$$\lambda = -\frac{\gamma}{2} \left(1 \pm \sqrt{1 - 4Q^2} \right). \quad (6)$$

The system is light damped if $\gamma < 2\omega_0$, because the solution still contains oscillating term ($\omega = \sqrt{\omega_0^2 - (\gamma/2)^2}$). The system is over-damped if $\gamma > 2\omega_0$ because the solution only contains exponential decays. Critical damping, which happens when $\gamma = 2\omega_0$ of great importance, because it is (generally) the fastest decaying solution,

$$x = (A + Bt)e^{-\gamma t/2}. \quad (7)$$

One should be able to tell whether an oscillation is over-damped, light-damped or critical-damped, and estimate its quality factor.

If the oscillator is driven by a force $F(x) = F_0 \cos(\omega t)$, the particular integral, which should be added to the general solution, is

$$x = \frac{f_0 \cos(\omega t)}{(\omega_0^2 - \omega^2) + i\gamma\omega}, \quad (8)$$

where $f_0 = F_0/m$. Normally, we define the response function $R(\omega)$ as

$$R(\omega) = \frac{x_0}{F_0} = \frac{1}{m[(\omega_0^2 - \omega^2) + i\gamma\omega]} \quad (9)$$

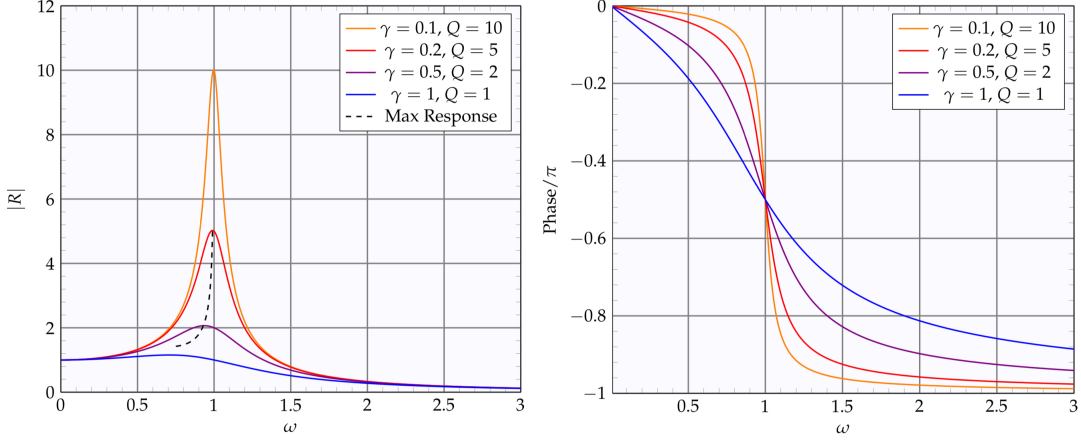


Figure 2: Response function of a driven oscillator, parameterised by different values of γ and Q .

which describes the displacement relative to the driving force. The modulus, $|R|$, gives the amplitude of the displacement per unit force, and the argument $\arg R$ gives the phase of the displacement with respect to the forcing sinusoid.

$$|R| = \frac{1}{m\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}}, \quad \arg R = \arctan\left(\frac{-\gamma\omega}{\omega_0^2 - \omega^2}\right). \quad (10)$$

In Fig. 2 it is clearly seen that at low frequencies, the motion is controlled by the force; at high frequencies, the motion is controlled by the inertia of the system. At $\omega_{\text{res}} = \omega_0\sqrt{1 - (1/2Q^2)}$, $|R|$ reaches its maximum, therefore we say that ω_{res} is the resonance frequency. At this frequency, $\arg R$ is roughly (a little smaller than) $-\pi/2$.

Note that, we didn't choose sinusoidal force by random, we chose it because every periodic function can be represented as its Fourier expansion, a sum of sine and cosines. The response function was defined because the particular integral is the steady-state solution to the equation of motion (exponential decays will vanish as time goes to infinity).

Similarly, we can derive the velocity's and acceleration's amplitude. Acceleration amplitude is proportional to the response function since $a = -\omega^2 x$, therefore it reaches maximum at $\omega_{a \text{ res}} = \omega_0/\sqrt{1 - (1/2Q^2)} = \omega_0^2/\omega_{\text{res}}$.

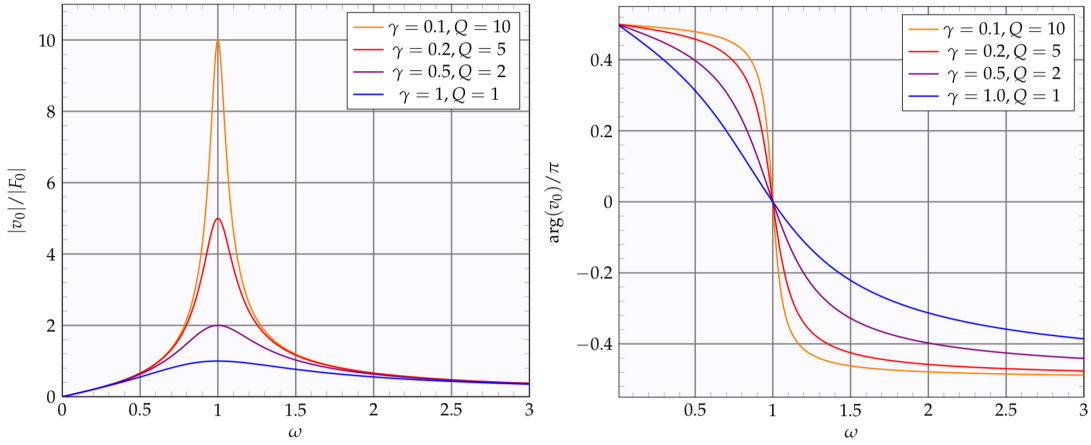


Figure 3: Amplitude and phase of velocity response, velocity $v = i\omega x$ has maximum at ω_0 independent of the damping.

The next task is to consider the power input of the driven force. Up to now, everything is in complex

form, but the imaginary parts have no physical meaning. Therefore, when we are calculating $P = Fv$, we need to take the real part first. However, this is a mathematical trick to avoid taking real part.

$$P = \Re(F)\Re(v) = \frac{1}{2}(F + F^*)\frac{1}{2}(v + v^*) = \frac{1}{2}\Re(Fv + Fv^*). \quad (11)$$

This only holds if F and v^* is commutative, but we don't need to worry about it here. Suppose the quantities both vary as $\exp(i\omega t)$, the power is

$$P = \Re(F)\Re(v) = \frac{1}{2}\Re(F_0 v_0 e^{2i\omega t} + F_0 v_0^*). \quad (12)$$

However, the first term will average to zero, the time-averaged power flowing into the system is simply $\langle P \rangle = \frac{1}{2}\Re(F_0 v_0^*)$. If there is a phase difference $\Delta\phi$ between force and velocity, the power will naturally be scaled by a factor of $\cos \Delta\phi$.

For a damped oscillator, the averaged power is simply $\frac{1}{2}b|v_0|^2$. Therefore power resonance occurs at frequency ω_0 independent of damping. In the power spectrum, bandwidth is defined by the full width of half maximum, which is simply $\Delta\omega = \gamma$. Therefore the quality factor can be equivalently defined by the resonance frequency divided by the bandwidth,

$$Q = \omega_0/\Delta\omega = \omega_0/\gamma. \quad (13)$$

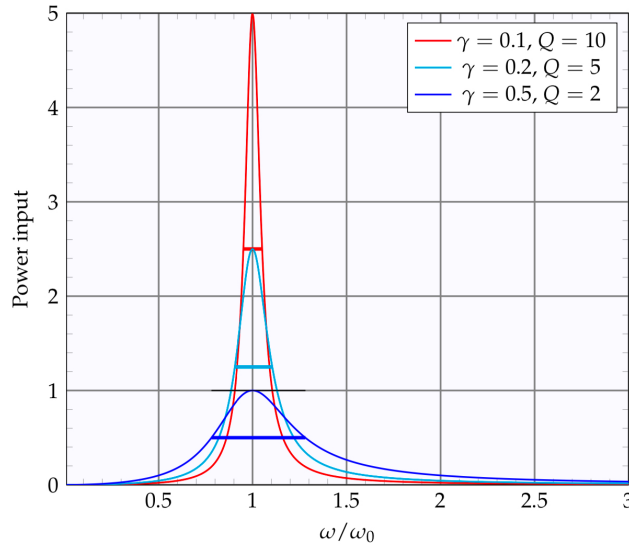


Figure 4: Power spectrum and bandwidth position.

Finally we need to define *impedance*. In any oscillation, impedance can be defined by amplitude of "force" (intensive property) divided by amplitude of "velocity" (time derivative of extensive property), here are some pairs: Table 1. In waves, impedance are defined identically, but we should always remember to add a minus sign before the impedance of a backward-traveling wave!

This may explain why the dimensions in electric circuits can relate to dimensions in mechanical systems³.

We shall skip anharmonic oscillations.

³I wish my junior high school teacher knew this!

Force	Displacement	Velocity
Force $\vec{F}$	Displacement $\vec{x}$	Velocity $\vec{v}$
Voltage V	Charge q	Current I
Pressure p	Volume V	dV/dt
Tension σ	Area S	dS/dt
Field times volume EV	Polarisation P	dP/dt
Field times volume $\mu_0 HV$	Polarisation M	dM/dt
EM wave field E	$\dots$	H

Table 1: Some pairs of general force and displacement.

1.1 Linear system at multiple frequencies

If we drive a system with different frequencies, the solution is found by adding the responses as if the individual driving forces act separately. This is mathematically straightforward, but slightly hard to visualise.

The resultant signal is particularly interesting if the forces are coherent (or close to coherent). To obtain a coherent oscillation, we typically split the signal from a single source and apply them to the system separately. The two signals will only have a phase difference, therefore the resultant oscillation is still an oscillation with the same frequency. Another way of introducing coherent signal is to couple similar sources and let them become phase-locked automatically.

Between different driving frequencies, we get the beating effect, because

$$\cos(\omega_1 t) + \cos(\omega_2 t) = 2 \cos\left(\frac{\omega_1 + \omega_2}{2}\right) \cos\left(\frac{\omega_1 - \omega_2}{2}\right). \quad (14)$$

The fast oscillation is at angular velocity $\frac{1}{2}(\omega_1 + \omega_2)$, but the energy, carried by the amplitude of the oscillation, is moving at an angular velocity $\frac{1}{2}(\omega_1 - \omega_2)$. If the two oscillations has different amplitude, the beating effect won't be that obvious, because some part of an oscillation cannot be canceled by the other.

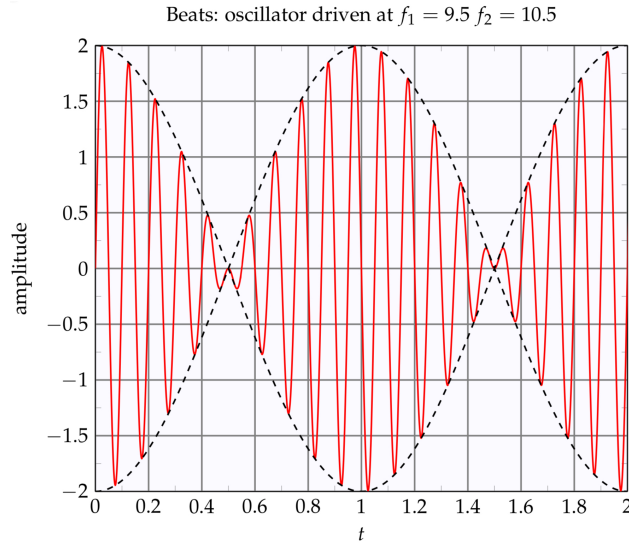


Figure 5: Beating effect. This is a good example to understand group velocity, which will be discussed in the following sections.

2 Waves

A wave moves energy and information from one point to another without bulk translation of the medium which it disturbs. The medium maybe a solid, liquid or a gas, or the field itself in the case of electromagnetic,

gravitational or quantum waves.

2.1 Wave equations

A wave can be characterised by a wavefunction $\psi = \psi(x, t)$, which describes the response at a set of space-time coordinates. There are two kinds of waves in the universe: transverse waves, whose oscillation is perpendicular to the direction of travel; longitudinal waves, whose oscillations is aligned with the direction of travel. They both result in the same form of wave equation,

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}, \quad (15)$$

where v is the (phase) velocity at which the (single-frequency) wave travels.

On a transverse string with constant tension T and mass per unit length λ , it is easy to prove that

$$v = \sqrt{\frac{T}{\lambda}}. \quad (16)$$

Through a media with Young's modulus E and density ρ , the velocity of longitudinal wave is

$$v = \sqrt{\frac{E}{\rho}}. \quad (17)$$

Of course, the form of wave equation implies that ψ can be expressed as a function of $x \pm vt$,

$$\psi(x, t) = \psi(x \mp vt, 0) = f(x \pm vt), \quad (18)$$

where the minus sign implying motion to plus x . If the velocity v remains constant, wave will propagate without changing its waveform. Also, the linearity of wave equation tells us different waves travels independently, and the total response must be the direct sum of each.

Using these interpretations, it is not hard to relate waves to Fourier expansion. And it's okay to say "any wave can be represented as a linear superposition of harmonic waves", where a harmonic wave is given by

$$\psi(x, t) = \Re \left(A e^{i(kx - \omega t)} \right), \quad k = \frac{\omega}{v} = \frac{2\pi}{\lambda}. \quad (19)$$

In higher dimensions, such as a drumskin, the wave equation becomes

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = \nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}. \quad (20)$$

where $v = \sqrt{\gamma/\sigma}$, γ is the tension per unit length, and σ is the area density. We have assumed that γ is a constant, while it is not generally.

Similarly, we shall change our representation of harmonic waves to general plane waves

$$\psi(\vec{x}, t) = \Re \left(A e^{i(\vec{k} \cdot \vec{x} - \omega t)} \right). \quad (21)$$

In the handout [1], the lecturer used another convention, to write phase as $(\omega t - kx)$. That convention does somehow help in optics, but I will stick to the current $(kx - \omega t)$ convention because I'm doing high energy physics.

Using the general Laplacian, it is convenient to write out all kinds of waves. For example, the wave equation for a spherical wave is

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}. \quad (22)$$

Consider a function $\psi(r, t) = f(r \pm vt)/r$, the LHS just gives $\nabla^2 \psi = f''/r$, the corresponding harmonic solution for outward propagating waves is

$$\psi(r, t) = \Re \left(\frac{A e^{i(kr - \omega t)}}{r} \right). \quad (23)$$

The $1/r$ dependence can be expected from energy conservation. For a cylindrical wave, the factor will naturally be $1/\sqrt{r}$ far-field⁴.

⁴Far field is necessary here, because the wave equation also gives $1/r^{5/2}$ terms. The full solution involves Bessel functions.

2.2 Polarisation

Transverse waves have polarisations because there are infinite number of directions in the same plane that are orthogonal to the direction of propagation. The waves with different polarisations are independent but have the same speed in the same medium.

Suppose the wave is traveling in the z direction. We can split the amplitude of a given wave into x and y components. The amplitude and relative phase of the displacement along x and y define the polarisation of the wave.

Consider a sine wave,

$$\begin{aligned}\psi_x &= A_x \cos(kx - \omega t); \\ \psi_y &= A_y \cos(kx - \omega t + \phi).\end{aligned}\tag{24}$$

If $\phi \neq 0$, the resultant displacement will rotate within the x - y plane, resulting in different kinds of polarisations. But generally, any polarisation can be expressed as the linear superposition of

- Linear polarisation, where $\phi = 0$; and
- Circular polarisation, where $\phi = (m + \frac{1}{2})\pi$. Note that we use the radio convention in this course: looking towards the wave source, right circular polarisation corresponds to clockwise motion. The IEEE convention is opposite.

People often call the polarisation with random ϕ an elliptical polarisation, but it can still be written as a linear superposition.

Mathematically we represent polarisations using Jones' vectors, and re-write Eq. 24 as

$$\begin{pmatrix} A_x \\ A_y e^{i\phi} \end{pmatrix} \Leftrightarrow \begin{pmatrix} A_x e^{-i\phi} \\ A_y \end{pmatrix}.\tag{25}$$

Using this expressions, a polariser would act as a Jones' matrix.

When wave is reflected by a plane, the polarisation states will change due to boundary conditions. For EM waves, refer to electromagnetism notes, and recall what we have learnt about the Brewster angle. Scattering also give rise to specific polarisation states. That's why polarisers are used in photography. Other uses of polarisation include liquid crystal screens, which was discussed in IA materials science course.

2.3 Energy in string wave

Energy conservation tells us that the energy input must equal to the transverse force times the transverse velocity. When dealing with oscillations, we have already figured out that

$$\langle P \rangle = \frac{1}{2} \Re(Fv^*) = \frac{1}{2} \Re(Z)|v|^2.\tag{26}$$

This must equal the energy per unit length of string times the wave speed.

The kinetic and potential energy per unit length are by definition

$$E_k = \frac{1}{2} \lambda \left(\frac{\partial \psi}{\partial t} \right)^2; \quad E_p = \frac{1}{2} T \left(\frac{\partial \psi}{\partial x} \right)^2,\tag{27}$$

which add up and average to $\frac{1}{2} \rho \omega^2 A_0^2$. The two energies do match.

2.4 Reflection and transmission

Waves reflect and transmits through a boundary. Suppose that we have two media, with impedance Z_1 and Z_2 , and normal incident. It is relatively easy to find the steady-state solution for harmonic waves⁵.

⁵It's hard to calculate the establishment process, though sometimes feasible, see problem sheet.

Since the solution is steady-state, the frequency at each side must be the same, but wavelength and wave speed are different. What we need to do is find the boundary conditions, which is the continuity of something, normally displacement and transverse force.

Suppose that we are lucky enough to know the displacement function of incident wave, the incident, reflected and transmitted waves can be written as

$$\begin{aligned} A_i e^{i(kx - \omega t)}; \\ r A e^{i(-kx - \omega t)}; \\ \tau A e^{i(kx - \omega t)}. \end{aligned} \quad (28)$$

Suppose $x = 0$ at the boundary, continuity will lead us to

$$1 + r = \tau; \quad Z_1(1 - r) = Z_2\tau, \quad (29)$$

which gives

$$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}; \quad \tau = \frac{2Z_1}{Z_1 + Z_2}. \quad (30)$$

The reflection coefficient of energy is naturally $R = |r|^2$, but be ware, $T = 1 - |r|^2 \neq |\tau|^2$.

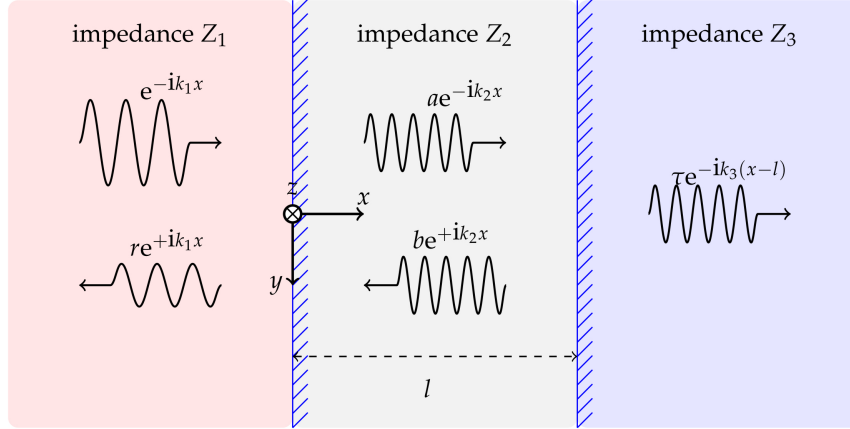


Figure 6: Impedance matching is done by inserting a very thin material to minimise reflection wave. When solving this problem, we often define $\gamma = e^{ik_2l}$ for convenience. This sets a further limit to Z_2 : $Z_2 = \sqrt{Z_1 Z_3}$, other than a quarter wavelength thickness $l = \lambda/4$.

2.5 Displacement wave

Our wave equation for sound waves was expressed in terms of the molecular displacements a but this is very hard to measure. We normally measure the pressure fluctuation with e.g. a microphone. We relate the two via

$$\psi_p = p - p_0 = \gamma p \frac{\partial a}{\partial x}. \quad (31)$$

If we assume a harmonic wave traveling to plus- x of the form

$$a = a_0 e^{i(kx - \omega t)}, \quad (32)$$

then $p - p_0 = i\gamma p k a$. i.e. the pressure leads the displacement by $\pi/2$ and has amplitude $\gamma p k a$, γ is the adiabatic constant of air.

The characteristic impedance Z , by definition, can be written as

$$Z = \frac{(p - p_0)\Delta S}{\dot{a}} = \frac{i\gamma p k a \Delta S}{i\omega a} = \Delta S \sqrt{\gamma p \rho}. \quad (33)$$

The factor $\sqrt{\gamma p \rho}$ is often referred as the acoustic impedance (impedance per unit area), $\mathcal{L}$.

The power carried by the acoustic wave is still calculated in the same manner,

$$I = \frac{1}{2} \Re(\psi_p \dot{a}^*), \quad (34)$$

which is the mean power per unit area, or intensity of the wave. An alternative expression is

$$I = \frac{1}{2} \mathcal{L} \omega^2 |a_0|^2. \quad (35)$$

To measure the intensity of sound wave, we normally use the dBA scale, A stands for amplitude,

$$\text{dBA level} = 10 \log_{10} \left(\frac{I}{I_{\text{ref}}} \right). \quad (36)$$

I_{ref} is the intensity that human can barely hear at 1 kHz, defined by an engineering value of

$$I_{\text{ref}} = \frac{p_{\text{ref}}^2}{\mathcal{L}} = \frac{(20 \mu\text{Pa})^2}{400 \text{ kgm}^{-2}\text{s}^{-1}} = 1 \times 10^{-12} \text{ Wm}^{-2}. \quad (37)$$

Human voice at one meter away is approximately sixty dB, which correlates to a power flux of ten micro-Watts.

For displacement waves in solids and liquids, we define the bulk modulus B as

$$dp = -B \frac{dV}{V}, \quad (38)$$

where $B = \gamma p$ for adiabatic gas. Solids are more complicated as they can carry shear stresses. Shear waves travel more slowly than compression waves and are important in earthquake analysis, because they lead to more damage. Normally, longitudinal waves travel faster than transverse ones, which has been investigated in the problem sheets.

2.6 Standing wave

Standing wave is a wave confined within a region of space by applying suitable boundary conditions. It is of great importance in musical instruments.

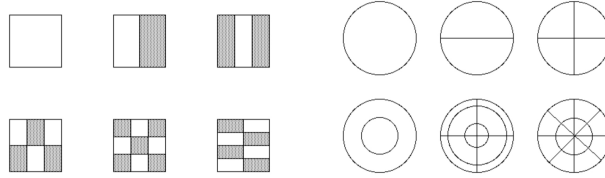


Figure 7: Lowest order standing wave modes of a rectangular and circular drumskin.

For 2D or 3D standing waves, we need to investigate its modes, from fundamental to high orders. Of course, different modes dominates at different frequencies.

2.7 Damped waves

Just like oscillators, waves lose energy through damping. A string wave will lose energy via air resistance, which normally has form $-\beta \dot{\psi} dx$. Suppose $\Gamma = \beta/\rho$, which expresses the damping, wave equation can be modified to

$$\frac{\partial^2 \psi}{\partial t^2} + \Gamma \frac{\partial \psi}{\partial t} = v^2 \frac{\partial^2 \psi}{\partial x^2}. \quad (39)$$

Plug in the harmonic wave solution $\psi = A \exp(i(kx - \omega t))$, we get

$$\omega^2 + i\Gamma\omega = v^2 k^2, \quad (40)$$

therefore k must in general be complex, which leads to exponential decays. Denote the real and imaginary parts of k by $k = k_r + ik_i$, we find that

$$\begin{aligned}\psi &= \exp(-k_i x) \Re[A \exp(i(kx - \omega t))]; \\ k_r^2 - k_i^2 &= \frac{\omega^2}{v^2}; \quad 2k_r k_i = \frac{\Gamma \omega}{v^2}.\end{aligned}\tag{41}$$

Of course, the phase speed is by definition ω/k_r , which is related to the damping!

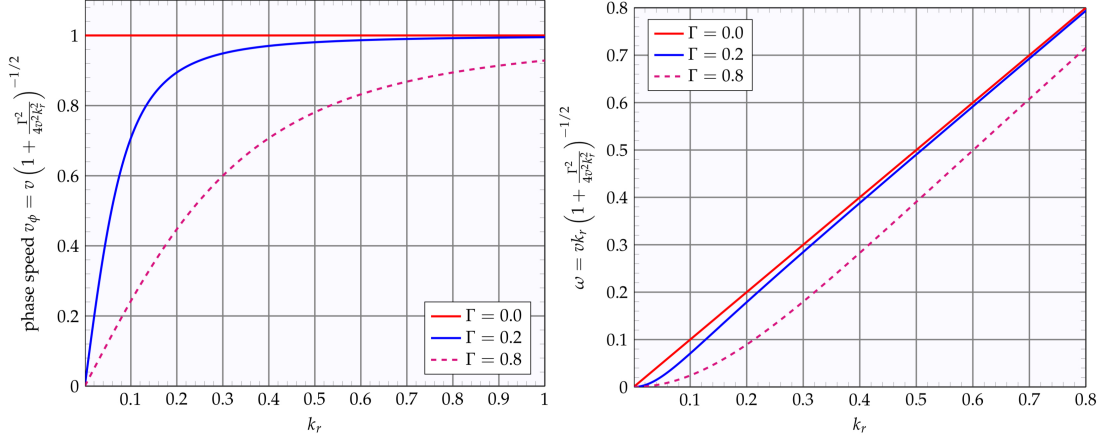


Figure 8: Dispersion of waves on a damped string, $v = 1$.

If $\Gamma \ll \omega$, the wave is light-damped, and $k_r \simeq \omega/v$. Therefore k_i can be estimated as $\Gamma/2v$.

If $\Gamma \gg \omega$, the wave is heavy-damped, $k^2 \simeq i\Gamma\omega/v^2$, therefore we have $k_r \simeq k_i \simeq \pm\sqrt{\Gamma\omega/2v^2}$. The wave decays over a short distance, and the decay length varies as $\omega^{-1/2}$.

The characteristic impedance of the damped string waves is, by definition

$$Z = -\frac{T\psi'}{\dot{\psi}} = \frac{Tk}{\omega} = \frac{T}{\omega}(k_r + ik_i).\tag{42}$$

For light damping and heavy damping, respectively

$$Z(\omega) = Z_0 \left(1 + \frac{i\Gamma}{2\omega}\right); \quad Z(\omega) = Z_0(1 + i)\sqrt{\frac{\Gamma}{2\omega}}.\tag{43}$$

Empowered with the impedance, we can easily calculate the reflection coefficient for light and heavy damping:

$$r(\omega) = -\frac{i\Gamma}{4\omega}; \quad r(\omega) = \frac{1 - (1 + i)\sqrt{\frac{\Gamma}{2\omega}}}{1 + (1 + i)\sqrt{\frac{\Gamma}{4\omega}}} \simeq -1.\tag{44}$$

2.8 Group velocity and dispersion

For a wave with no dispersion, we don't need to care about group velocity because it is simply the same as phase velocity⁶. However, things are different with dispersion. In the damped waves section we have encountered the first dispersion relation, where ω was expressed in terms of k . The group velocity is defined to be

$$v_g = \frac{d\omega}{dk},\tag{45}$$

which can be understood as the velocity at which amplitude (and therefore energy) travels. The difference between phase and group velocities leads to dispersion effects of wave packets, which was examined in the computing question.

⁶Or, in another way of understanding, has no definition.

Guided waves and electromagnetic waves are discussed in electromagnetism course, we shall skip them here.

3 Physical Optics

From Maxwell's theory of electromagnetism, we know that light is in fact electromagnetic waves. In this section we will be playing with them, but only in classical situations (relatively large dimensions, and large number of photons). Also, for simplicity we will ignore polarisation and assume that light beams of the same frequency are all coherent.

3.1 Huygens' principle and general diffraction

Diffraction arises from Huygens' principle. In the first chapter of his famous book *Traité de la Lumière* (Treatise on Light), he wrote⁷:

Ily a encore à confiderer dans l'émanation de ces ondes, que chaque particule de iamatiere, dans laquelle une onde s'étend, ne doit pas communiquer fôn mouvement feulement à la particule prochaine, qui elldans la ligne droite tirée du point lumineux mais qu'elle en donne auflî neceffairement à toutes les autres qui la touchent, & qui s'oppofent à fon mouvement.

This indicates that he himself, and some of other scientists around the world, has realised that light interacts with something else. At that time, it was called aether. Now, it has a more suitable name: field.

Huygens' discussion was indeed purely qualitative⁸, and everything was done on paper using geometrical methods. But his method worked, in the following chapters his method explained almost every optical effect that was observed.

In modern English, Huygens' principle may be written as

Each point on a wavefront acts as a source of secondary wavelets which propagate, overlap, interfere, and thus carry the wavefront forward.

Or, in other words:

Each point on a wavefront acts as a source of secondary spherical waves, the new wavefront will be the envelope of the secondary waves.

I prefer the latter expression, and I wish the lecturer told it to everyone. However, it is clear that there are two envelopes, traveling forwards and backwards, that both satisfy Huygens' principle. Fresnel fixed this by adding an *obliquity* or *inclination factor*, which describes the fall-off in intensity of the wavelets with angle θ , away from the forward direction. The factor is

$$K(\theta) = \frac{1 + \cos \theta}{2}, \quad (46)$$

to get excellent agreement to the experimental data. In addition, Fresnel found that the secondary wavelets have an amplitude $(-i/\lambda)$ relative to the primary wave. The extended form of Huygens' principle is known as Huygens-Fresnel theory.

So far we have figured out how light propagate and interfere. But how to mathematically describe it?

Suppose we have a diffraction aperture Σ , which is built inside the aperture plane. We can build a coordinate system, x - y , and consider an element of aperture (dx, dy) at (x, y) . What we need to do is describe:

- The propagation from source to the aperture.
- Interfere at the aperture.

⁷There are direct discussions on wavefronts, but personally this sentence is more important.

⁸Reads like philosophy in a modern physicist's perspective.

- Propagation from aperture to observer.

Let's look at the first step, propagation from source to the aperture. Suppose the source produces spherical waves with a strength a_S , and is at a distance s from the element of aperture. The wave arriving at the aperture element is therefore

$$\psi_1(\vec{r}) = \frac{a_S e^{iks}}{s}. \quad (47)$$

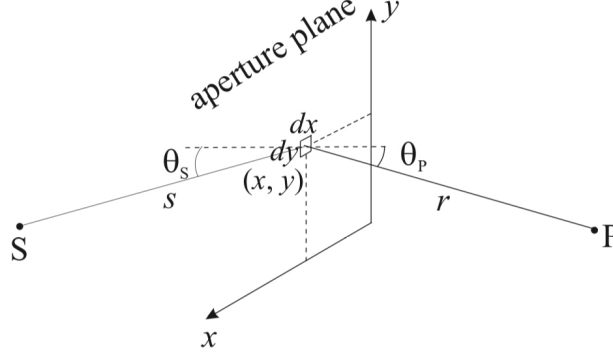


Figure 9: General case of diffraction.

The aperture can change the amplitude and phase of the incident radiation, and for a planar aperture we can describe its transmission properties by the complex aperture function $h(x, y)$. Usually, it is a combination of top-hat functions. The element of aperture can now be considered to act as a source of secondary wavelets with a strength and phase given by

$$a_\Sigma = A\psi_1(x, y)h(x, y) \, dx \, dy. \quad (48)$$

The amplitude of the wavelets relative to the incoming wave is $A = -i/\lambda$. The secondary wave generates a disturbance at the observer, which is a distance r away from the aperture element

$$d\psi_p = -\frac{i}{\lambda} \frac{a_S e^{iks}}{s} h(x, y) \, dx \, dy \, K(\theta) \frac{e^{ikr}}{r}. \quad (49)$$

It turns out that $K(\theta)$ depends on both θ_S and θ_P ,

$$K = \frac{\cos \theta_S + \cos \theta_P}{2}. \quad (50)$$

Often, the angles are assumed to be small and we shall set $K \simeq 1$ for convenience.

The total amplitude at point P should be a sum of diffraction pattern throughout the aperture plane,

$$\psi_p = \int_{\Sigma} -\frac{i}{\lambda} h(x, y) K(\theta_S, \theta_P) \frac{a_S e^{ik(s+r)}}{sr} \, dx \, dy. \quad (51)$$

Note that both s and r are functions of x and y . This is known as the Fresnel-Kirchhoff diffraction integral.

3.2 Fraunhofer diffraction

Although Fresnel diffraction gives the general equation of the diffraction pattern if the assumptions are satisfied, it is very hard to compute without a digital computer. However, with some further assumptions we can turn the equation into a simpler one that is doable by hand, and preserves its physical value at the same time.

Assume $s \rightarrow \infty$ so that waves are at normal incidence to the aperture, and we denote the coordinates of point P in the observer's plane by (x_0, y_0) . Also, we will only consider P close to the axis, where x_0/L and y_0/L are small enough to assume $K = 1$.

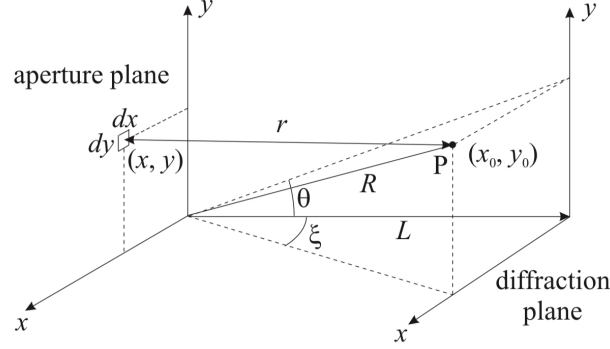


Figure 10: Geometry of Fraunhofer diffraction.

The distance r is therefore given by

$$r = [L^2 + (x_0 - x)^2 + (y_0 - y)^2]^{1/2} \simeq R - \frac{x_0 x + y_0 y}{R} + \frac{x^2 + y^2}{2R}, \quad (52)$$

where $R^2 = L^2 + x_0^2 + y_0^2$. The phase difference is proportional to $k\Delta r$, we will further assume that x/R and y/R are also small enough so that the second-order terms vanish. By doing this the Fresnel diffraction integral turns into Fraunhofer diffraction integral:

$$\psi_p = \int_{\Sigma} \psi_{\Sigma} h(x, y) \exp \left[-ik \left(\frac{x_0 x + y_0 y}{R} \right) \right] dx dy, \quad (53)$$

with $\psi_{\Sigma} = \text{const}$ according to our assumptions. We have dropped the pre-factor because it makes no sense to know the absolute intensity. Only relative values matter. Note that we get Fraunhofer diffraction only if

$$\frac{k(x^2 + y^2)}{2R} = \frac{kr^2}{2R} \ll \pi, \quad (54)$$

as discussed previously. Re-arranging the equation, this condition sets a minimum value of R :

$$R \gg \frac{D^2}{\lambda} \quad (55)$$

where D is the maximum value of r for the aperture. Since x and y are independent variables, we can just look at them separately. Note that $y = R \sin \theta$, if we define $q = k \sin \theta$, we have

$$\psi_p \propto \int_{\Sigma} h(y) e^{-iqy} dy \propto \mathcal{F}[h(y)], \quad (56)$$

which tells us the famous result:

The amplitude distribution for a Fraunhofer diffraction is proportional to the Fourier transform of the aperture function.

Thanks to the linearity of Fourier transform and the convolution theorem, we can always divide an arbitrary aperture into several basic ones, and derive the Fraunhofer diffraction pattern separately.

The power spectrum is naturally proportional to the square of amplitude spectrum.

Impulse response functions Suppose we strike a linear system with a delta-like impulse, the response is its impulse response function. Since everything convolved with delta function is itself, the general response of the system to a force $F(t)$ is

$$x(t) = \mathcal{F}^{-1}[R(\omega)\mathcal{F}[F(t)]], \quad (57)$$

where $R(\omega)$ is the impulse response function in Fourier space.

Now we are able to calculate any kind of Fraunhofer diffraction, the rest of this section is simply experimental instructions.

Original space	Fourier space
Delta function	$e^{-i\omega t_0}/\sqrt{2\pi}$
Deltas with interval Δx	Deltas with interval $2\pi/\Delta x$
Top-hat function with full width Δx	Sinc function with full zero-order maxima $2\pi/\Delta x$
Gaussian of t	Gaussian of ω
Decaying exponential	Low-pass filter $1/\sqrt{2\pi}(1 + i\omega)$

Table 2: Commonly-used Fourier transforms.

Conditions for observing Fraunhofer diffraction We have made the assumption that $R \gg D^2/\lambda$, but how can we achieve it in laboratories? We can use lenses on either side of the aperture to convert the diverging beams into parallel beams, and vice versa. The observer plane must be the focus plane of the second lens.

Natural line width Emission of light from lights gives a discrete spectrum, because the energy levels of electron in atoms are fixed. However, due to perturbations and uncertainty principle of energy $\Delta E \Delta t \sim \hbar$, each line still have a natural width of about $\Delta\omega \sim 1/\Delta t \sim 1 \times 10^8 \text{ rad s}^{-1}$. Therefore the power spectrum has a Lorentzian line shape rather than the ideal delta function.

Collision broadening Collision broadening is also known as pressure broadening, because microscopic collisions results in macroscopic pressure. It arises from the collisions between atoms while they are emitting light. This limits the coherence of the emitted light waves. The mean time between collisions in a gas is $\tau \sim 1/n\Sigma u$, where n is the number density, Σ is the collision cross section, and u is the average speed. This will give a broadening effect of about $\Delta\omega \sim 1 \times 10^{10} \text{ rad s}^{-1}$.

Doppler broadening Also known as thermal broadening. The movement of atoms causes Doppler effect (small, but still measurable). The observed frequency of an atom moving at speed u is

$$\omega = \omega_0 \sqrt{\frac{1 + \frac{u}{c}}{1 - \frac{u}{c}}} \simeq \omega_0 \left(1 + \frac{u}{c}\right). \quad (58)$$

The Boltzmann distribution tells us the line shape of intensity spectrum is

$$I(\omega) \propto \exp\left(-\frac{mc^2(\omega - \omega_0)^2}{2\omega_0^2 k_B T}\right), \quad (59)$$

which gives a standard deviation of $\sim 4 \times 10^9 \text{ rad s}^{-1}$ in air at room temperature.

Note that Lorentzian line is a bit flatter than Gaussian.

Grating resolution and Rayleigh Criterion Lord Rayleigh concluded that the condition of resolving two Gaussian peaks is that they are separated by a full width of half maxima⁹. In modern mathematics, this can be done by limiting a single maximum of the summed intensity.

Resolution of Optical Instruments In physical optics, the finite circular extent of the lens creates diffraction. An incoming plane wavefront from a distant point object will produce an *Airy disc* pattern in the image plane. The angular radius is

$$\alpha = \frac{1.22\lambda}{D} \quad (60)$$

where D is the diameter of the objective lens/mirror.

⁹Or, the maxima of one peak sits on the first zero of the other.

Babinet's principle If two aperture functions satisfy

$$h_1(x, y) + h_2(x, y) = 1, \quad (61)$$

the sum of their Fraunhofer diffraction patterns will also give one (pure bright).

3.3 Fresnel diffraction

The diffraction integral becomes a Fourier transform far-field, but observing closer to the aperture, finer structure appears.

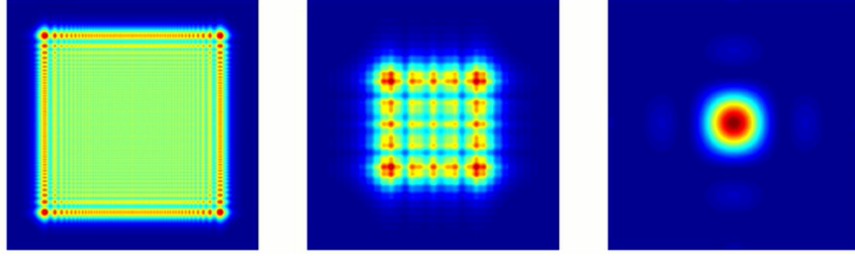


Figure 11: Diffraction from a square aperture, near, close, and far.

For distance of order D^2/λ or smaller, we must include the quadratic and possible higher-order terms in the phase of the integral. This is Fresnel diffraction¹⁰.

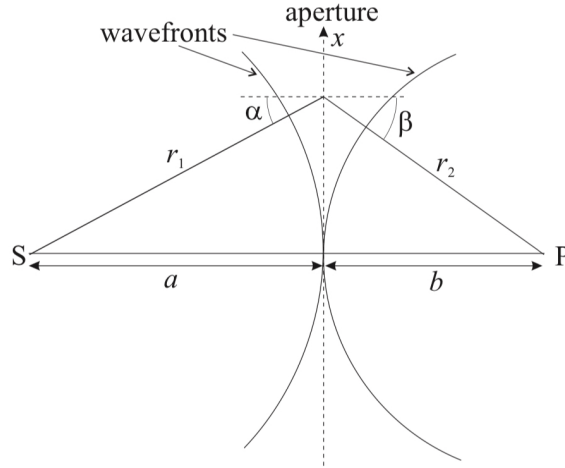


Figure 12: Geometry of Fresnel diffraction

We now assume that the angles to the edge of the aperture are small enough that we can neglect the obliquity factor, and thus take $K(\theta) = 1$. Also, we assume that the variations in r_1 and r_2 over the aperture are negligible as far as the denominator is concerned: $r_1 \sim a$ and $r_2 \sim b$; but note variations in $(r_1 + r_2 - a - b)$ are not negligible compared with λ , and so have a significant effect on the phase term.

Under these assumptions, the diffraction integral becomes

$$\psi_p = \int_{\Sigma} h(x, y) \exp\left(\frac{ik(x^2 + y^2)}{2R}\right) dx dy. \quad (62)$$

¹⁰Professor Xihua Zhong's book [2] explains this part a lot more clearly than our handout. But sadly the book was originally written in Chinese, which is not understood by everyone.

For a rectangular aperture, the integral is

$$\psi_p \propto \int_{x_1}^{x_2} \exp\left(\frac{ikx^2}{2R}\right) dx \int_{y_1}^{y_2} \exp\left(\frac{iky^2}{2R}\right) dy, \quad (63)$$

and it can be rearranged into

$$\int_{u_1}^{u_2} \exp\left(\frac{i\pi u^2}{2}\right) du \int_{v_1}^{v_2} \exp\left(\frac{i\pi v^2}{2}\right) dv. \quad (64)$$

Define the Fresnel integrals

$$C(w) = \int_0^w \cos\left(\frac{\pi u^2}{2}\right) du; \quad S(w) = \int_0^w \sin\left(\frac{\pi u^2}{2}\right) du, \quad (65)$$

the result is therefore the vector from $C(w_1) + iS(w_1)$ to $C(w_2) + iS(w_2)$. The numerical integration solution is visualised in Fig. 13.

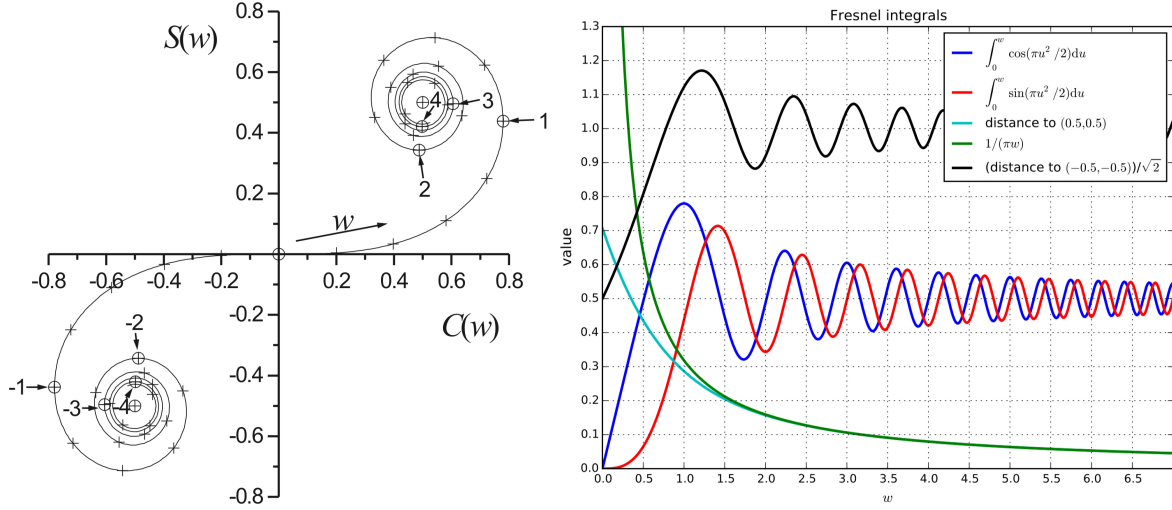


Figure 13: Cornu spiral (left) and Fresnel integrals (right). Together, they tell us what happens if the aperture is a slit that gradually opens.

Half-period zones If we restrict our aperture to a small region at the centre, it can work as a lens and give us very high intensity by coloring the half-period zones. In this region, the exponential convergence of Fresnel integrals can be ignored and we simply have an oscillating amplitude with radius r .

$$\psi_p \propto \int_0^{s=r_a^2} \frac{\pi K(s)}{(a^2 + s)^{1/2}(b^2 + s)^{1/2}} \exp\left(\frac{i\pi s}{\lambda R}\right) ds, \quad s = \rho^2. \quad (66)$$

In other words, looking at the equation, we are not interested in the pre-factor, but we are going to use the exponential because it is a direct indication of oscillation.

The first Fresnel half-period zone is defined to be $0 \leq \phi(\rho) \leq \pi$, which corresponds to $\rho^2 \leq \lambda R$. And the n -th zone is in the range

$$\sqrt{(n-1)\lambda R} \leq \rho \leq \sqrt{n\lambda R}. \quad (67)$$

Odd numbered zones can sum to very high intensity but it will (if n gets really large) converge to a certain value.

3.4 Interference

Nothing to add, refer to the CPhO outline.

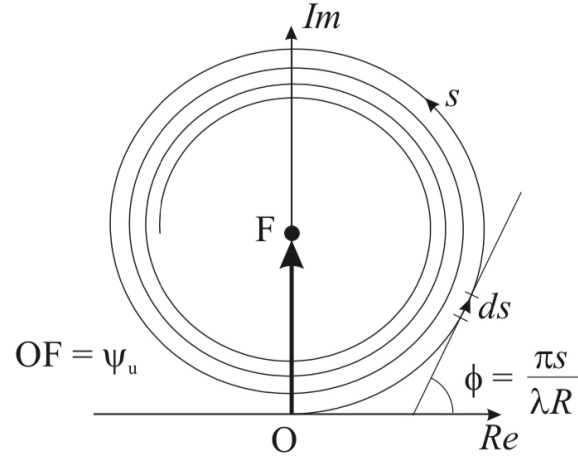


Figure 14: Phasor diagram of Fresnel diffraction of a circular aperture.

References

- [1] T. Euser et al. *Oscillations, Waves and Optics Handout*. University of Cambridge, 2024.
- [2] X. Zhong. *Modern Foudamentals of Optics*. Peking University Press, 2nd edition, 2012.